LEED v4 for BD+C: Schools

Project Checklist

Project Name: Brecksville-Broadview Hgts. New Elementary School

Y	?	N				
	1		Credit	Integrative Process		1

1	0	12	**Location and Transportation**			15
		x	Credit	LEED for Neighborhood Development Location - develop within the boundary of certified LEED for Neighborhood Development		15
		1	Credit	Sensitive Land Protection - develop land that has been previously developed		1
		2	Credit	High Priority Site - develop at a historic site, priority designation, or brownfield remediation		2
		5	Credit	Surrounding Density and Diverse Uses - develop site with req'd density within 1/4 mile, or req'd uses within 1/2 mile		5
		4	Credit	Access to Quality Transit - develop mass transit served site or site with >50% of stundents walking		4
		1	Credit	Bicycle Facilities - must not be located near 10 approved ammenities		1
		1	Credit	Reduced Parking Footprint - cannot provide additional parking above code mimimum		1
1			Credit	Green Vehicles - must provide electric charging for 2% of parking	$32,000.00 a.	1

7	3	2	**Sustainable Sites**			12
Y			Prereq	Construction Activity Pollution Prevention - conform to requirements of 2012 US EPA and CGP	In project	Required
Y			Prereq	Environmental Site Assessment - ESA Phase 1 required, Phase 2 if needed. Must remediate if contamination exists.	In project	Required
1			Credit	Site Assessment - conduct site survey to include (topography, hydrology, climate, vegetation, soils, human use, human health effects)	$15,000.00	1
2			Credit	Site Development - Protect or Restore Habitat - preserve and protect 40% of greenfields on site	In project	2
1			Credit	Open Space - open outdoor space ≥30% of the total site area	In project	1
	3		Credit	Rainwater Management - 95th Percentile, replicate natrual hydrology, manage on-site the runoff from development	$250,000.00 b.	3
2			Credit	Heat Island Reduction - provide plant shading, paving w/ SR≥0.28, and roof w/ SR≥64 (low slope) and SR≥32 (steep slope)	$1,100,000.00 c.	2
		1	Credit	Light Pollution Reduction - meet uplight and light trespass requirements		1
		1	Credit	Site Master Plan - must meet 4 SS specific credits and must have future planned development		1
1			Credit	Joint Use of Facilities - make 3 specific building spaces open to the general public	In project d.	1

5	0	7	**Water Efficiency**			12
Y			Prereq	Outdoor Water Use Reduction - no irrigation or 30% reduced irrigation	In project	Required
Y			Prereq	Indoor Water Use Reduction - reduce water consumption by 20% of baseline	In project	Required
Y			Prereq	Building-Level Water Metering - water meters that measure total potable water use	In project	Required
2			Credit	Outdoor Water Use Reduction - no irrigation required or 50% LWR from the calculated baseline	In project	2
2		5	Credit	Indoor Water Use Reduction - reduce water consumption by 30% of baseline, compliant washing machines, compliant kitchen equip.	In project	7
		2	Credit	Cooling Tower Water Use		2
1			Credit	Water Metering - provide subsystem water metering	$60,000.00 e.	1

18	5	8	**Energy and Atmosphere**			31
Y			Prereq	Fundamental Commissioning and Verification	$98,000.00 f.	Required
Y			Prereq	Minimum Energy Performance - whole building energy simulation. Demonstrate 5% improvement from baseline	$30,000.00 g.	Required
Y			Prereq	Building-Level Energy Metering - provide builidng level energy meters	In project	Required
Y			Prereq	Fundamental Refrigerant Management - do not use CFCs	In project	Required
5		1	Credit	Enhanced Commissioning - system and envelope commissioning	$37,000.00 h.	6
12		4	Credit	Optimize Energy Performance - whole building energy simulation	$46,000.00	16
				Optimize Energy Performance - chilled beam mechanical system	$2,405,000.00 i.	
				Optimize Energy Performance - additional mechanical rooms (2,000 sf)	$500,000.00 j.	
				Optimize Energy Performance - additional mechanical and electrical engineering	$115,000.00	
1			Credit	Advanced Energy Metering - metering at whole building sources and individual sources that represent >10% annual consumption	$54,000.00	1
	1	1	Credit	Demand Response - requires contract with the power company to control building systems remotely	$50,000.00	2
	3		Credit	Renewable Energy Production - 1pt for 1%, 2pt for 5%, 3pt 10% of total building usage. potentially buy renewable energy credits 10 y	Annual Cost	3
		1	Credit	Enhanced Refrigerant Management - use no refrigerants		1
	1	1	Credit	Green Power and Carbon Offsets	$6,000.00	2

a. 8 electric vehicle charging stations and signage
b. Would likely require permeable paving at all paved areas and a larger detention basin for the roof generated rain water.
c. Use concrete paving in lieu of asphalt - approximately $4/sf. No cost for roof.
d. The cafeteria, gymnasium, one classroom, and joint parking would be shared with the City.
e. Meter water use at plumbing fixtures and all HVAC equipment and tie into the building BAS system
f. $0.53/sf
g. cost for energy modeling
h. Premium for enhanced commissioning
i. Current HVAC system is estimated at $26.77/sf. Chilled beam system is approximately $35-$40/sf. $8/sf - $13/sf add
j. 2,000 sf x $250/sf
k. Contractor additional operational hours to process and track LEED submittals

6	0	7	**Materials and Resources**				13
Y			Prereq	Storage and Collection of Recyclables	In project		Required
Y			Prereq	Construction and Demolition Waste Management Planning	In project		Required
		5	Credit	Building Life-Cycle Impact Reduction			5
2			Credit	Building Product Disclosure and Optimization - Environmental Product Declarations	In project		2
1		1	Credit	Building Product Disclosure and Optimization - Sourcing of Raw Materials	$20,000.00	k.	2
1		1	Credit	Building Product Disclosure and Optimization - Material Ingredients	$20,000.00	k.	2
2			Credit	Construction and Demolition Waste Management	$20,000.00		2

9	2	5	**Indoor Environmental Quality**			16
Y			Prereq	Minimum Indoor Air Quality Performance	In project	Required
Y			Prereq	Environmental Tobacco Smoke Control	In project	Required
Y			Prereq	Minimum Acoustic Performance	In project	Required
2			Credit	Enhanced Indoor Air Quality Strategies - entryway system, cross-contamination prevention, filtration	In project	2
2	1		Credit	Low-Emitting Materials - comply with 3-6 of the 8 categories	In project	3
1			Credit	Construction Indoor Air Quality Management Plan - meet or exceed requirements of SMACNA for occupied buildings under construction	In project	1
2			Credit	Indoor Air Quality Assessment - flush out mechanical system prior to occupancy	In project	2
1			Credit	Thermal Comfort - design building HVAC systems and envelope to meet the requirements of ASHRAE 55-2010	In project	1
1	1		Credit	Interior Lighting - lighting control and lighting quality	In project	2
		3	Credit	Daylight - demonstrate 55%, 75%, or 90% spatial daylight autonomy		3
		1	Credit	Quality Views - 75% of occupied floor area shall meet 2 of the 4 kinds of views		1
		1	Credit	Acoustic Performance - hvac sysstems less than 35 dBA, learning spaces STC per ANSI S12.60-2010, windows STC ≥ 35		1

1	1	4	**Innovation**			6
	1	4	Credit	Innovation	In project	5
1			Credit	LEED Accredited Professional	In project	1

1	1	2	**Regional Priority**			4
1			Credit	Regional Priority: Specific Credit - *Enhanced Indoor Air Quality Strategies*		1
	1		Credit	Regional Priority: Specific Credit - *Rainwater Management*		1
		1	Credit	Regional Priority: Specific Credit		1
		1	Credit	Regional Priority: Specific Credit		1

48	13	47	**TOTALS**	$4,858,000.00	110

Certified: 40 to 49 points, **Silver:** 50 to 59 points, **Gold:** 60 to 79 points, **Platinum:** 80 to 110